

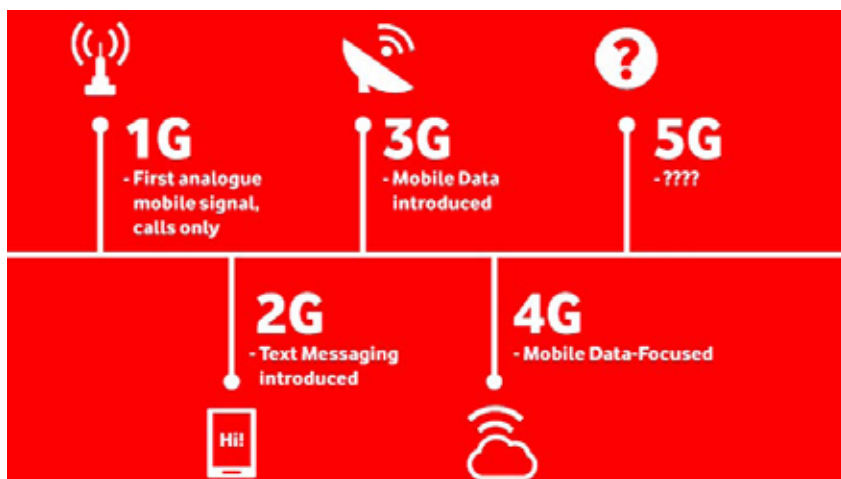


Figure 1: Mobile data speed (Credits: <https://www.techindulge.com/technovation/1g-2g-3g-4g-and-5g-wireless-phone-technology-explained-meaning-and-differences/>)

high-definition mobile TV, video conferencing, 3D television, and cloud computing. The maximum speed of a 4G network when the device is moving is 100Mbps. The speed is 1Gbps for low-mobility communication such as when the caller is stationary or walking.

5G: 5G promises significantly faster data rates, higher connection density, much lower latency, and energy savings, among other improvements. It offers data transfer rates of up to 20Gbps, and allows users to download ultra-high-definition videos and access the internet at lightning-fast speeds. It also offers lower latency, better network coverage, and improved call quality.

Wireless carriers

Wireless carriers are cellular technology companies that provide mobile telecommunications services for supported devices.

CDMA: CDMA stands for code division multiple access. It is a handset-specific technology. CDMA is not very common, and is available in comparatively fewer carriers and countries. These devices are exclusive to Canada, Japan, and the United States.

The CDMA technology does not support transmission of voice and data simultaneously. It is faster, provides better security and comes with built-in encryption.

GSM: The GSM (global system for mobile) standard was launched by AT&T in Finland. Every device uses SIM (subscriber identity module) to communicate with the provider network.

GSM is highly available and globally used. Over 80% of the entire world's mobile networks use it. It uses time-division multiple access (TDMA) and frequency-division multiple access (FDMA). GSM supports the transmission of both voice and data at once. It is slower, less secure, and has no default encryption compared to CDMA.

LTE: Long-term evolution (LTE) is used for faster data transfer and higher capacity. Different variations of LTE networks exist across carriers that use different frequencies. For example, Sprint, T-Mobile, Verizon, and AT&T all have their own bands of LTE.

LTE moves large packets of data to an internet protocol system (IPS). Old methods like CDMA and GSM only transferred small amounts of data.

Both LTE and 4G evolved together, with LTE being the industry standard for the forward edge of the fourth generation's advancement.

4G LTE functionality has two key preconditions: a network that supports the ITU-R (ITU Radiocommunication Sector) standard speeds, and a device powerful enough to match and handle

the speeds of that network.

Both GSM and CDMA have switched to LTE as the global 4G standard.

SATCOM: For users who lack traditional landline or cellular coverage, satellite communication (SATCOM) is an option for mobile device use. It uses an artificial satellite to transmit radio signals covering far greater distances and wider areas than most other radio technologies.

Satellite phones function in remote locations as they do not rely on phone transmission lines or cellular towers. Most satellite phones have limited connectivity to the internet, and data rates tend to be slow, but they come with GPS capabilities to indicate the user's position in real time.

Mobile management

Mobile management is all about keeping your devices and data safe — from mobile device management (MDM) that sets access policies to mobile content management (MCM), which secures data.

Mobile device management

(MDM): Managing mobile device access and usage in an organisation is a security challenge, where company-owned or user-owned mobile devices are centrally managed. This includes setting policies on apps, camera usage, data, access, and so on. It also focuses on access control features such as screen locks, multi-factor authentication, remote wipe and geofencing.

Mobile content management

(MCM): Mobile content management secures the content present on mobile devices; it involves monitoring and restrictions on file sharing, online content viewing, and uploading.

It also entails centralised data management using solutions such as Microsoft Office 365. It implements data loss prevention (DLP) to prevent any sensitive information leakage, encrypts data on the device, and restricts and blocks external or removable drives.

Mobile application management

(MAM): Managing mobile apps is challenging, as not all applications are